The Power of AI in Medical Imaging

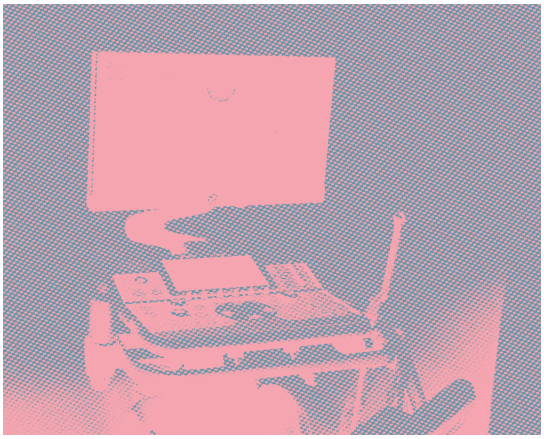
Podcast

Summary

In this episode, the conversation focuses on how artificial intelligence is being used to improve medical imaging. Guest Jacob Jaremko, a radiologist and biomedical engineer, reflects on his path from engineering to medicine and explains how his early work with neural networks eventually connected to his current efforts in radiology and AI.

Jaremko describes radiology as a field where interpretation still contains a great deal of variability. Different experts may look at the same image and describe it differently, and even the idea of a single "ground truth" can be difficult to define. In medical imaging, what appears in a scan may be open to interpretation, and even pathology findings can depend on how experts classify what they see. Because of this, he argues that there is value in making image analysis more standardized, reproducible, and linked more closely to real patient outcomes over time.

A major part of the episode focuses on ultrasound. Unlike CT, MRI, or X-ray systems, ultrasound can be portable and brought directly to the patient. Jaremko explains why this makes ultrasound especially important for health equity, since it can potentially reach people in remote communities or people who are not well served by conventional care settings. At the same time, ultrasound is difficult to interpret because the images are generated from sound waves and often look unclear to untrained users. This creates an opportunity for AI to support both image acquisition and interpretation.



He outlines two main ways AI can help with ultrasound. First, AI can guide users while they are capturing images, helping them identify whether they are scanning the right anatomy and whether the image quality is good enough. Second, AI can assist with interpretation by identifying anatomy, flagging possible abnormalities, and offering a screening-level assessment. In this way, AI does not remove the need for experts, but can lower the expertise required for early screening and help determine which patients actually need follow-up with specialists.

The discussion uses infant hip dysplasia as a key example. Jaremko explains that ultrasound can identify this condition early in life, when treatment is simple and highly effective. But in some remote settings, families may need to travel long distances just to access a short scan. He describes how AI-assisted ultrasound tools could allow nurses, local community members, or other non-specialist users to perform initial scans closer to home, with only the abnormal cases being referred onward for specialist confirmation and treatment. This has the potential to reduce travel burdens while improving access to early diagnosis.

Finally, Jaremko reflects on the challenge of moving AI from the lab into real clinical settings. He discusses the balance of being a scientist, clinician, and co-founder, and emphasizes his goal of using AI-enabled tools to make care more effective, efficient, and equitable in practice.

Key Terms

Term	Definition
Deep Learning	A type of machine learning that uses layered neural networks made up of connected nodes, often called artificial neurons, which process information and adjust their connections during training to learn patterns in data such as medical images.
Deep Reinforcement Learning	An approach that combines deep learning with reinforcement learning to support decision-making across a sequence of steps.
Ground Truth	The reference answer used for training or evaluation, though the podcast emphasizes that in medicine this can be difficult to define and may itself involve interpretation.
Health Equity	The goal of making health care access and outcomes more fair and more available across different populations and settings.
Hip Dysplasia	A condition where the hip joint does not form properly, which can lead to later problems if it is not detected and treated early.
Image Guidance	AI support that helps a user capture the right images by indicating whether the anatomy and image quality are adequate.
Medical Imaging	Techniques used to look inside the body, including ultrasound, MRI, CT, and X-ray.
Portable Ultrasound	Small ultrasound devices that can be brought directly to the patient rather than requiring the patient to travel to a major imaging centre.
Segmentation	The process of identifying and outlining important structures within a medical image.
Triage	The process of sorting cases so that people who most need expert follow-up are prioritized.
Ultrasound	An imaging method that uses sound waves to create images of structures inside the body.

3 Key Takeaways

1. Ultrasound becomes far more powerful when AI helps people capture and interpret it.
2. Medical imaging is not always as objective or standardized as people assume.
3. AI can support more equitable care by helping bring screening closer to the patient.

Discussion Questions

1. Medical imaging is often viewed as highly precise, yet the podcast emphasizes that experts can interpret the same scan differently. Where in your own field do people assume there is one clear answer, even though interpretation or judgment still plays a major role?

2. Jaremko describes ultrasound as a technology that can travel to the patient instead of forcing the patient to travel to the technology. What other services, tools, or systems in your context could become more equitable if they were redesigned around portability or access?

3. The podcast presents AI as a way to lower the expertise needed for early screening, while still keeping experts involved for more complex cases. In your own work, where could AI or automation handle first-pass triage so that specialists can focus on the cases where their expertise matters most?

Relevant Resources

For more information on topics mentioned in this podcast, check out:

- For learners interested in Jacob Jaremko's research and industry work:
  - Jaremko's Researcher Profile Page
  - Jaremko's Personal Website
  - Publications
- On medical imaging and ultrasound:
  - Radiological Society of North America (RSNA)
  - Stanford AIMI Center – Artificial Intelligence in Medicine and Imaging

References

Alberta Machine Intelligence Institute. (2025, April 15). The power of AI in medical imaging | Approximately Correct podcast [Video]. YouTube. https://youtu.be/hKvnbjET04I?si=9FQGIs0xAe9VPSyd

amii.ca | hello@amii.ca

Alberta Machine Intelligence Institute
1101, 10065 Jasper Ave NW
Edmonton, AB T5J 3B1 Canada

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